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HT Board Booster

Class X Science

Physics

CBSE Board Paper Question Bank

Chapter Wise Questions

Ch - 4
Magnetic Effects of Electric Current

Session 2026-27



Class X Science Physics
Ch- Magnetic Effect of Electric Current
Previous Year CBSE Board Question Paper

CBSE BP 2025-26
(Set- 31/1/1- 31/5/1)
Section-A (one Marker Questions)

Q.1. **Assertion (A):** Direction of force exerted on a current-carrying conductor placed in a magnetic field remains same if the directions of both magnetic field and current flowing through the conductor are reversed. (Set- 31/2/1)
Reason (R): The direction of force exerted on a current-carrying conductor placed in an external magnetic field can be determined by using magnetic compass.

Q.2. **Assertion (A):** The needle of a compass, kept in an external magnetic field, always aligns itself in the North-South direction of the Earth. (Set- 31/3/1)
Reason (R): The behaviour of the needle of the compass is similar to the behaviour of a freely suspended bar magnet.

Q.3. **Assertion (A):** With the help of compass, magnetic field lines can be traced outside the bar magnet. (Set- 31/4/1)
Reason (R): The north pole of a compass needle always points towards the south pole of the magnet when brought close to it.

Q.4. **Assertion (A):** The needle of a magnetic compass kept in strong external magnetic field, always aligns itself in north-south direction on the earth. (Set- 31/5/1)
Reason (R): Behaviour of the needle of a compass is same as that of a freely suspended bar magnet.

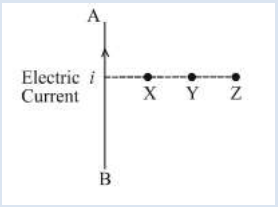
Section-D (four Marker Questions)

Q.5. (a) Describe an activity to show that a current carrying conductor, placed in an external magnetic field experiences a force. (Set- 31/1/1)

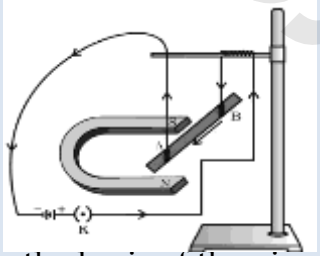
(b) Imagine that you are sitting in a chamber with your back to one wall.
An electron beam, moving horizontally towards the front wall from the back wall, is deflected by a strong magnetic field to your right side. Find the direction of the magnetic field.

Q.6. (a) The pattern of magnetic field due to a current carrying wire depends upon the shape made by that wire. Justify.

(b) A current carrying straight wire AB is shown in the given diagram. Out of X, Y and Z on which point will the strength of magnetic field be maximum and why? (Set- 31/1/1)



Q.7.



On the basis of the given circuit arrangement, answer the following questions: (Set- 31/4/1)

(a) What happens to the rod AB when the key K is closed and why?

(b) Name and state the rule that determines this behaviour of the rod AB in the above situation.

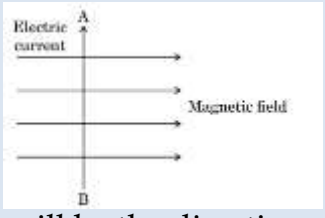
Section-E (five Marker Questions)

Q.8. Attempt either (a) or (b) : (Set- 31/2/1)

(a) (i) State the rule which gives the direction of force acting on a current-carrying conductor placed in a magnetic field.

(ii) What is a fuse? Why is it called a safety device?

(iii) Why are electrical appliances with metallic bodies connected to the mains through a three-pin plug, whereas an electric bulb is connected with a two-pin plug?

(iv) 

What will be the direction of force experienced by the conductor AB due to the magnetic field, when current flows in the direction shown ? Name the law you applied to find the direction.

OR

(b) (i) Draw the pattern of magnetic field lines through and around a current-carrying circular loop.

(ii) State the rule to find out the direction of the magnetic field inside and around the circular loop.

(iii) Give reasons:

(I) Why should a burnt-out fuse be replaced by another fuse of identical rating only, in an electric circuit?

(II) Why is soft iron-core used in making an electromagnet?

Q.9. Attempt either (a) or (b) : (Set- 31/3/1)

(a) (i) What is a solenoid? Draw a diagram showing the magnetic field pattern due to a current-carrying solenoid (inside and outside).

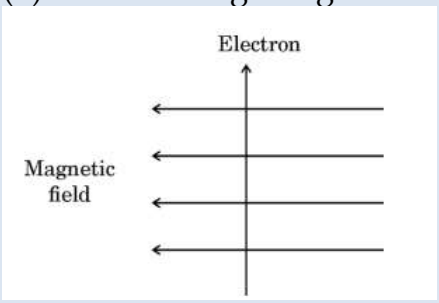
(ii) Name any two factors on which the magnitude of the magnetic field due to a current-carrying solenoid depends.

(iii) A current-carrying solenoid is stretched and converted into a straight wire carrying current. Draw a diagram showing the magnetic field pattern around it and mark the direction of the magnetic field.

OR

(b) (i) With the help of a diagram, show that a current-carrying conductor placed in an external magnetic field, experiences a force. Indicate the direction of the field, current and force clearly.

(ii) The diagram given below shows an electron moving in a uniform magnetic field.



(I) What will be the direction of the force acting on the electron?

(II) What will happen to the direction of the force acting on the electron, if the direction of the magnetic field as well as the direction of motion of the electron, both are reversed?

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Q.10. (i) Draw a diagram showing the magnetic field lines around a bar magnet. Write any two characteristics of magnetic field lines. (Set- 31/5/1)

(ii) Write the statement of Right-hand thumb rule.

(iii) The concentric circles representing the magnetic field around a current carrying straight wire become larger and larger as we move away from the wire. How does magnetic field change on these concentric circles?

OR

(i) Draw a diagram, showing the magnetic field lines in and around the current carrying solenoid. Write any two salient features of these lines.

(ii) What is an electromagnet and how is it made?

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CBSE BP 2024-25 (Set- 31/1/1- 31/6/1)

Section-A (one Marker Questions)

Q.11. **Assertion (A):** No two magnetic field lines are found to cross each other. (Set- 31/1/1)
Reason (R): The compass needle cannot point towards two directions at the point of intersection of two magnetic field lines.

Q.12. Which one of the following statements is not true about a bar magnet? (Set- 31/3/1)
(a) It sets itself in north-south direction when suspended freely.
(b) It has attractive power for iron filings.
(c) It produces magnetic field lines.
(d) The direction of magnetic field lines inside a bar magnet is from its north pole to its south pole.

Q.13. The strength of magnetic field produced inside a long straight current carrying solenoid does not depend upon: (Set- 31/3/1)
(a) number of turns in the solenoid
(b) direction of current flowing through the solenoid
(c) material of the core filled inside the solenoid
(d) radius of the coil of the solenoid

Q.14. **Assertion (A):** Magnetic field lines around a bar magnet never intersect each other.
Reason (R): Magnetic field produced by a bar magnet is a quantity that has both magnitude and direction. (Set- 31/3/1)

Q.15. **Assertion (A):** The pattern of the magnetic field of a solenoid carrying a current is similar to that of a bar magnet. (Set- 31/5/1)
Reason (R): The pattern of the magnetic field around a current carrying conductor is independent of the shape of the conductor.

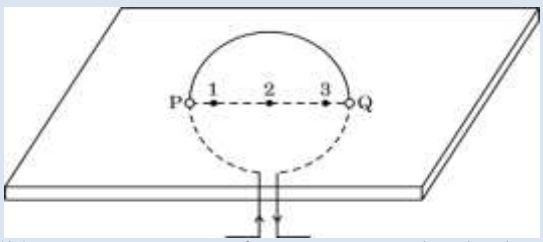
Q.16. **Assertion (A):** In the common domestic circuits the earth wire is connected to a metallic plate buried deep inside the earth. (Set- 31/6/1)
Reason (R): Earth wire ensures that any leakage of current to the metallic body of the appliance keeps its potential to that of the earth, so the user may not get a severe electric shock.

Section-B

Q.17. What are magnetic field lines. List two important properties of magnetic field lines. (Set- 31/4/1)

Section-C

Q.18. Consider a rectangular cardboard having two holes P and Q through which a current carrying circular loop has been inserted as shown in the diagram. (Set- 31/3/1)
(a) Make this diagram on your answer sheet and draw three magnetic field lines, one each passing through the points 1 (near P), 2 (at the centre of the loop) and 3 (near Q).



(b) List two factors on which the intensity of the magnetic field produced at the centre of the loop depends.
(c) Name the rule you will apply to determine the direction of magnetic field produced due to a current carrying straight conductor.

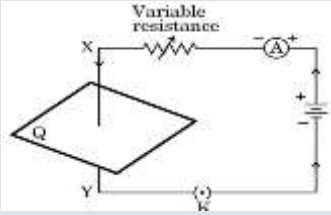
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Q.19. (a) "The third wire of earth connection is very important in domestic electric appliances." Justify this statement. (Set- 31/5/1)

(b) List two precautions to be taken to avoid the overloading of domestic electric circuits.

Section-D

Q.20. The given figure shows the current passing through the straight conductor XY. (Set- 31/2/1)



(i) Copy the diagram and draw the magnetic field lines when current flows from conductor X to Y.

(ii) Name and state the rule used in determining the direction of the magnetic field lines in the situation given above.

(iii) State Fleming's left hand rule. Using this rule, determine the direction of force applied on an electron entering a uniform magnetic field as shown in the figure.



OR

(i) Define the term solenoid. Draw the pattern of the magnetic field lines in and around a current carrying straight solenoid. Mark on the pattern the (i) direction of current,

(ii) direction of field lines near the ends of the solenoid, and

(iii) region where the magnetic field is uniform.

(iv) How would you make an electromagnet using a current carrying solenoid?

Q.21. (i) Draw the pattern of the magnetic field lines for the two parallel straight conductors carrying current of same magnitude "I' in opposite directions as shown. Show the direction of magnetic field at a point O which is equidistant from the two conductors. (Consider that the conductors are inserted normal to the plane of a rectangular cardboard.) (Set- 31/4/1)

(ii) In our houses we receive A.C. electric power of 220 V. In electric iron or electric heater cables having three wires with insulation of three different colours red, black and green are used to draw current from the mains.

(a) What are these three different wires called? Name them colourwise.

(b) What is the potential difference between the red wire and the black wire?

(c) What is the role of the wire with green insulation in case of accidental leakage of electric current to the metallic body of an electrical appliance?

OR

(i) By using the given experimental set-up. How can it be shown that:

(a) a force is exerted on the current-carrying conductor AB when it is placed in a magnetic field.

(b) the direction of force can be reversed in two ways.

(i) When will the magnitude of the force be highest?

(ii) State Fleming's left hand rule.

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Q.22. What are magnetic field lines? How is the direction of magnetic field at a point determined? Draw the pattern of magnetic field lines of the magnetic field produced by a current carrying circular loop. Mark on it the direction of (i) current and (ii) magnetic field lines.
Name the two factors on which the magnitude of the magnetic field due to a current carrying coil depends. (Set- 31/6/1)

OR

Why can't two magnetic field lines cross each other? Draw magnetic field lines showing the direction of the magnetic field due to a current carrying long straight solenoid. State the conclusion which can be drawn from the pattern of magnetic field lines inside the solenoid. Name any two factors on which the magnitude of the magnetic field due to this solenoid depends.

Section-E

Q.23. In our homes, we receive the supply of electric power through a main supply also called mains, either supported through overhead electric poles or by underground cables. In our country the potential difference between the two wires (live wire and neutral wire) of this supply is 220 V. (Set- 31/1/1)

(a) Write the colours of the insulation covers of the line wires through which supply comes to our homes.

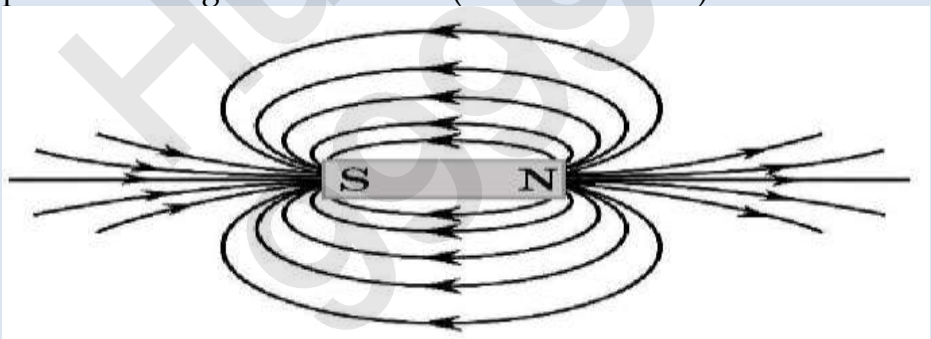
(b) What should be the current rating of the electric circuit (220 V) so that an electric iron of 1 kW power rating can be operated?

(c) (i) What is the function of the earth wire? State the advantage of the earth wire in domestic electric appliances such as electric iron.

OR

(c) (ii) List two precautions to be taken to avoid electrical accidents. State how these precautions prevent possible damage to the circuit/appliance.

Q.24. In order to obtain magnetic field lines around a bar magnet, a student performed an experiment using a magnetic compass and a bar magnet. The magnet was placed on a sheet of white paper fixed on a drawing board. Using magnetic needle he obtained on the paper a pattern of magnetic field lines (as shown below) around the bar magnet.



(a) By convention, the field lines emerge from north pole and merge at south pole. Why? Give reason.

(b) State the relationship between strength of the magnetic field and the degree of closeness of the field lines.

(c) (A) (i) No two field lines can ever intersect each other. Give reason.
(ii) The magnetic field in a given region is uniform. Draw a diagram to represent it.

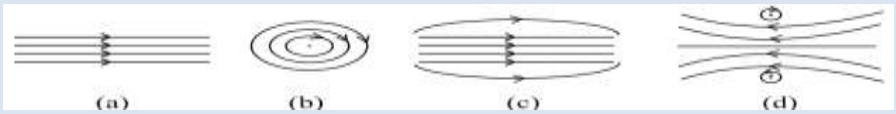
OR

(c) (B) Draw the pattern of the magnetic field lines through and around a current carrying solenoid. What does the pattern of field lines inside the solenoid represent?

CBSE BP 2023-24 (Set- 31/1/1- 31/5/1)

Section-A (one Marker Questions)

Q.25. The pattern of the magnetic field produced inside a current carrying solenoid is: (Set- 31/1/1)

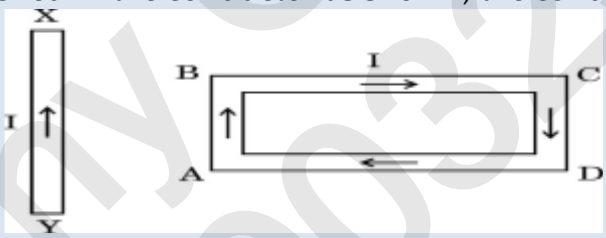


Q.26. Strength of magnetic field produced by a current carrying solenoid **DOES** NOT depend upon: (Set- 31/2/1)

(a) number of turns in the solenoid
(b) direction of the current flowing through it
(c) radius of solenoid
(d) material of core of the solenoid

Q.27. **Assertion (A) :** The deflection of a compass needle placed near a current carrying wire decreases when the magnitude of an electric current in the wire is increased.
Reason (R) : Strength of the magnetic field at a point due to a current carrying conductor increases on increasing the current in the conductor. (Set- 31/2/1)

Q.28. A rectangular loop ABCD carrying a current I is situated near a straight conductor XY, such that the conductor is parallel to the side AB of the loop and is in the plane of the loop. If a steady current I is established in the conductor as shown, the conductor XY will:



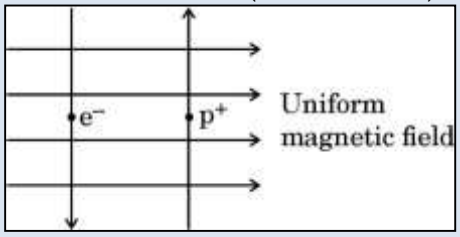
(a) Remain stationary. (Set- 31/4/1)
(b) Move towards the side AB of the loop.
(c) Move away from the side AB of the loop.
(d) Rotate about its axis.

Q.29. **Assertion (A) :** Magnetic field lines never intersect each other. (Set- 31/4/1)
Reason (R) : If they intersect, then at the point of intersection, the compass needle would point towards two directions, which is not possible.

Q.30. The current carrying device which produces a magnetic field similar to that of a bar magnet is : (Set- 31/5/1)

(a) A straight conductor (b) A circular loop
(c) A solenoid (d) A circular coil

Q.31. A uniform magnetic field exists in the plane of paper as shown in the diagram. In this field, an electron (e -) and a positron (p +) enter as shown. (Set- 31/5/1)



(a) both pointing into the plane of the paper.
(b) both pointing out of the plane of the paper.
(c) pointing into the plane of the paper and out of the plane of the paper respectively.
(d) pointing out of the plane of the paper and into the plane of the paper respectively.

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SECTION B

Q.32. (i) Two magnetic field lines do not intersect each other. Why? (Set- 31/1/1)

(ii) How is a uniform magnetic field in a given region represented? Draw a diagram in support of your answer.

Q.33. Draw a labelled diagram to show the pattern of magnetic field lines produced due to a current carrying straight conductor. Mark on it the direction of current in the conductor and the direction of magnetic field lines. (Set- 31/3/1)

OR

Name the device used to magnetise a piece of magnetic material.

Draw a labelled diagram to show the arrangement used for the magnetisation of a cylinder made of soft iron.

Section – C

Q.34. Name and state the rule to determine the direction of a: (Set- 31/1/1)

(i) Magnetic field produced around a current carrying a straight conductor.

(ii) Force experienced by a current carrying straight conductor placed in a magnetic field which is perpendicular to it.

Q.35. Draw a diagram to show the pattern of magnetic field lines on a horizontal sheet of paper due to a straight conductor passing through its centre and carrying current vertically upwards. Mark on it (i) the direction of current in the conductor and (ii) the corresponding magnetic field lines. State right hand thumb rule and check whether the directions marked by you are in accordance with this rule or not. (Set- 31/2/1)

Q.36. A student fixes a sheet of white paper on a drawing board. He places a bar magnet in the centre of it. He sprinkles some iron filings uniformly around the bar magnet. Then he taps the drawing board gently and observes that the iron filings arrange themselves in a particular pattern. (Set- 31/4/1)

(a) Why do iron filings arrange in a particular pattern?

(b) What does the crowding of iron filings at the ends of the magnet indicate ?

(c) What do the lines, along which the iron filings align, represent?

(d) If the student places a cardboard horizontally in a current carrying solenoid and repeats the above activity, in what pattern would the iron filings arrange ? State the conclusion drawn about $CaSO_4 . \frac{1}{2} H_2O$. the magnetic field based on the observed pattern of the lines.

Q.37. (a) What happens when a bundle of wires of soft iron is placed inside the coil of a solenoid carrying a steady current ? Name the device obtained. Why is it called so ? (Set- 31/5/1)

(b) Draw the magnetic field lines inside a current carrying solenoid. What does this pattern of magnetic field lines indicate?

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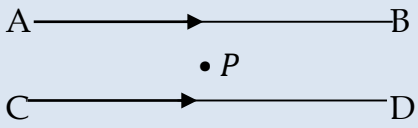
CBSE BP 2022-23 (Set- 31/1/1- 31/6/1)

Section-A (one Marker Questions)

Q.38. In domestic electric circuits the wiring with 15 A current rating is for the electric devices which have: (Set- 31/1/1)

(a) Higher power ratings such as geyser.
(b) Lower power ratings such as fan.
(c) Metallic bodies and low power ratings.
(d) Non-metallic bodies and low power ratings.

Q.39. The resultant magnetic field at point 'P' situated midway between two parallel wires (placed horizontally) each carrying a steady current I is (Set- 31/1/1)



(a) in the same direction as the current in the wires.
(b) in the vertically upward direction.
(c) zero
(d) in the vertically downward direction.

Q.40. **Assertion (A):** The strength of the magnetic field produced at the centre of a current carrying circular coil increases on increasing the number of turns in it. (Set- 31/1/1)
Reason (R): The current in each circular turn has the same direction and the magnetic field due to each turn then just adds up.

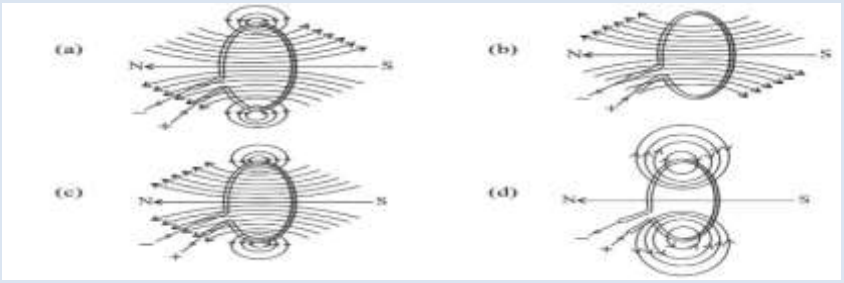
Q.41. A positron enters a uniform magnetic field at right angles to it as shown. The direction of force experienced by the positron will be: (Set- 31/2/1)



(a) to the right (b) to the left (c) into the page (d) out of the page

Q.42. **Assertion (A):** Magnetic field lines do not intersect each other. (Set- 31/2/1)
Reason (R): Magnetic field lines are imaginary lines, the tangent to which at any point gives the direction of the field at that point.

Q.43. The correct pattern of magnetic field lines of the field produced by a current carrying circular loop is: (Set- 31/4/1)



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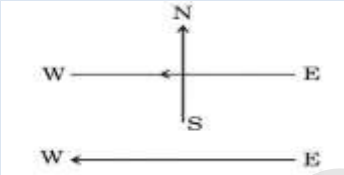
Q.44. An alpha particle enters a uniform magnetic field as shown. The direction of force experienced by the alpha particle is: (Set- 31/4/1)



(a) towards right
(b) towards left
(c) into the page
(d) out of the page

Q.45. **Assertion (A):** A current carrying straight conductor experiences a force when placed perpendicular to the direction of magnetic field. (Set- 31/4/1)
Reason (R): The net charge on a current carrying conductor is always zero.

Q.46. A constant current flows in a horizontal wire in the plane of the paper from east to west as shown in the figure. The direction of the magnetic field will be north to south at a point:



(a) Directly above the wire. (b) Directly below the wire. (Set- 31/5/1)
(c) Located in the plane of the paper on the north side of the wire.
(d) Located in the plane of the paper on the south side of the wire.

Q.47. For a current in a long straight solenoid, N and S poles are created at the two ends. Among the following statements, the incorrect statement is : (Set- 31/5/1)

(a) The magnetic field lines inside the solenoid are in the form of straight lines, which indicates that the magnetic field is uniform at all points inside the solenoid.
(b) The strong magnetic field produced inside the solenoid can magnetize the soft iron placed inside it.
(c) The pattern of the magnetic field associated with a current carrying solenoid is different from the pattern of the magnetic field around a bar magnet.
(d) The N and S poles exchange positions when the direction of current through the solenoid is reversed.

Q.48. **Assertion (A)**: Two magnetic field lines around a current carrying straight wire do not intersect each other. (Set- 31/5/1)
Reason (R): The magnitude of the magnetic field produced at a given point increases as the current through the wire increases.

Q.49. The magnetic field inside a long straight current carrying solenoid: (Set- 31/6/1)

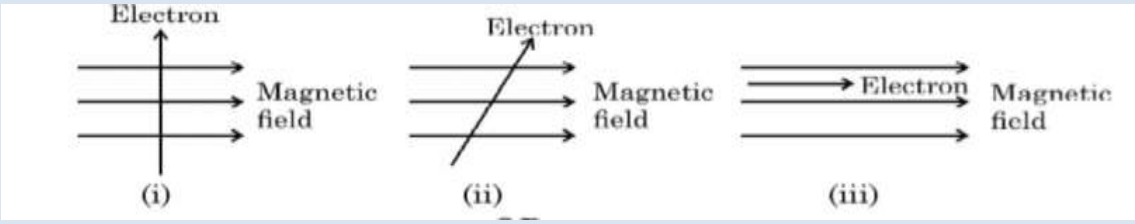
(a) is zero. (b) decreases as we move towards its end.
(c) increases as we move towards its end. (d) is same at all points.

SECTION-C

Q.50. (a) (i) State the rule used to find the force acting on a current carrying conductor placed in a magnetic field. (Set- 31/1/1)

(ii) Given below are three diagrams showing entry of an electron in a magnetic field. Identify the case in which the force will be (1) maximum and (2) minimum respectively. Give reason for your answer.

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(1) a current carrying solenoid
(2) a bar magnet

(ii) List two distinguishing features

Q.51. Explain the meaning of overloading of an electrical circuit. List two possible causes due to which overloading may occur in household circuits. Write one preventive measure that should be taken to avoid overloading of domestic circuits. (Set- 31/2/1)

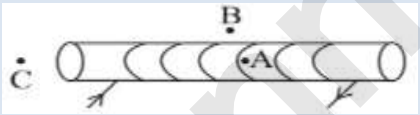
Q.52. (A) (i) Why is an alternating current (A.C.) considered to be advantageous over direct current (D.C.) for the long distance transmission of electric power? (Set- 31/4/1)

(ii) How is the type of current used in household supply different from the one given by a battery of dry cells?

(iii) How does an electric fuse prevent the electric circuit and the appliances from a possible damage due to short circuiting or overloading.

OR

(B) For the current carrying solenoid as shown, draw magnetic field lines and give reason to explain that out of the three points A, B and C, at which point the field strength is maximum and at which point it is minimum ?



Q.53. (a) (i) A straight cylindrical conductor is suspended with its axis perpendicular to the magnetic field of a horse-shoe magnet. The conductor gets displaced towards left when a current is passed through it. What will happen to the displacement of the conductor if the (Set- 31/5/1)

(1) Current through it is increased?
(2) horse-shoe magnet is replaced by another stronger horse-shoe magnet?
(3) Direction of current through it is reversed?

(ii) Name and state the rule for determining the direction of force on a current carrying conductor in a magnetic field.

OR

(b) Draw the pattern of the magnetic field produced around a vertical current carrying straight conductor passing through a horizontal cardboard. Mark the direction of current and the magnetic field lines. Name and state the rule which is used to determine the direction of magnetic field associated with a current carrying conductor.

Q.54. What is a solenoid? When does a solenoid behave as a magnet? Draw the pattern of the magnetic field produced inside it showing the directions of the magnetic field lines. (Set- 31/6/1)

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